CIS 4060 Android Programming Labs

Android Programming: The Big Nerd Ranch Guide (2022)

This document explains the semester-long Android programming lab assignment. It is separate from your semester project and runs on its own schedule, alongside it.

1. Assignment Overview

Throughout the semester, you will work through the assigned tutorials from Android Programming: The Big Nerd Ranch Guide (2022), following the schedule in Section 5. Each assigned chapter is a lab, due by the end of the week shown on the schedule.

For each assigned chapter, complete the following:

1. Complete the tutorial as written in the chapter.
2. Complete at least one end-of-chapter Challenge. You may complete additional Challenges for extra practice, but only one is required.
3. Commit and push all source code for the chapter to your Git repository.
4. Complete a Lab Report for the chapter and commit it alongside your code. See Section 3.

2. Android API Changes

Android development moves quickly, and APIs often change after a textbook goes to print. Some examples in the book may not work exactly as written by the time you reach them. This is expected. It is not a sign you have done something wrong.

You are expected to troubleshoot these issues on your own: consult current official documentation, search for updated guidance, or use other reliable sources. You may also use AI tools to help you troubleshoot.

This is the point. Successfully adapting older code to a current Android API is an expected part of this assignment, not an obstacle to it. It is also exactly the kind of problem-solving real Android developers do constantly, since the platform never stops changing.

3. The Lab Report

For every chapter, complete a short Lab Report using the separate Lab Report template, and commit it to your repository alongside that chapter's code. It records which Challenge you completed and, when relevant, how you worked through an API or SDK adaptation issue.

If the chapter went smoothly, mark the report “No issues encountered” and you are done; there is nothing else to fill in.

If you ran into an issue caused by an API or SDK change, document it briefly on the report:

- What caused the issue
- How you resolved it
- Where you found the solution (official documentation, a website, Stack Overflow, AI, or another source)

Keep it short. About one short paragraph or a few bullet points is enough.

If you used AI. Submit a PDF of the complete AI conversation along with your Lab Report for that chapter. This applies any time AI contributed to resolving an adaptation issue.

4. Submission Requirements

By the end of each lab week:

- Commit and push all source code, tutorial and Challenge, for every chapter assigned that week.
- Complete and commit a Lab Report for each chapter assigned that week, as described in Section 3.

See the Tentative Schedule in Course Information for specific due dates.

5. Grading

Each chapter is worth 20 points, for up to 600 points across the semester. This is separate from your semester project grade.

You earn the full 20 points for a chapter by submitting both your code and your Lab Report for that chapter. Missing either one means the points for that chapter are not earned.

The exact quality of your solution may vary. As long as your code is neat and your solution is generally correct, you earn the points. This is not graded on a sliding scale of quality, and small differences in how you solved a problem do not cost you points.

6. Semester Schedule

These are the assigned chapters for the semester, with the lab week each one is due.

Table 1. *Semester lab schedule*

Lab Week	Chapter	Title
W1	1	Your First Android Application
W1	2	Interactive User Interfaces
W2	3	The Activity Lifecycle
W2	4	Persisting UI State
W3	5	Debugging Android Apps
W3	6	Testing
W4	7	Your Second Activity
W4	8	Android SDK Versions and Compatibility
W5	9	Fragments
W5	10	Displaying Lists with RecyclerView
W6	11	Creating User Interfaces with Layouts and Views

Lab Week	Chapter	Title
W6	12	Coroutines and Databases
W7	13	Fragment Navigation
W7	14	Dialogs and DialogFragment
W9	15	The App Bar
W9	16	Implicit Intents
W9	17	Taking Pictures with Intents
W9	18	Localization
W10	19	Accessibility
W10	20	Making Network Requests and Displaying Images
W11	21	SearchView and DataStore
W11	22	WorkManager
W12	23	Browsing the Web and WebView
W12	24	Custom Views and Touch Events
W14	25	Property Animation
W14	26	Introduction to Jetpack Compose
W14	27	UI State in Jetpack Compose
W14	28	Showing Dialogs with Jetpack Compose
W16	29	Theming Compose UIs
W16	30	Afterword